

JESS Thermodynamic Database v8.9

Ruthenium

24-Jan-23

Reaction No. 4866							
RuO4(s) + e-1 = RuO4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.9(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:1.00(3SF)	0	CRV	6330
Note (2): Assumed wrong phase; see RuO4 + e-1 = RuO4-1							

Reaction No. 4867							
RuO4-1 + e-1 = RuO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.59(0.01SD)	6	MIS	775
2	25	0	Inf. Dilution	E:0.59(2SF)	5	CRV	6330

Reaction No. 5042							
Ru+3_Cl-1(2) + e-1 = Ru+2_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Cl-1 salt	lgK:-0.2(0.1SD)	5	MPL	854

Reaction No. 5044							
Ru+3_Cl-1(2) + e-1 = Ru+2_Cl-1 + Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1.5:6.8	HCl	lgK:1.4(0.05SD)	5	MEF	853

Reaction No. 5048							
Ru+3 + e-1 = Ru+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.2(0.05SD)	5	MEF	854
2	25	1	H2SO4	lgK:3.0(0.05SD)	5	MEF	859
3	25	0	Inf. Dilution	E:0.2487(4SF)	5	CRV	6330
4	25	0	Inf. Dilution	E:0.23(2SF)	0	ENS	7276
5	25	0.1	HClO4	E:0.11(2SF)	0	MEF	867
6	25	0	Inf. Dilution	dH:10.1(0.1SD)	5	MEF	854

Reaction No. 5049							
Ru+3_Cl-1(3) + e-1 = Ru+2_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Cl-1 salt	lgK:-2.(0.8SD)	3	MPL	854

Reaction No. 5050							
Ru+3_OH-1(3)_(s) = Ru+3 + 3<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:-36.(1.SD)	3	MSL	860
2	25	0.1:2	Unknown	lgK:-34.(1.SD)	3	MSL	861

Reaction No. 5051							
Cl-1 + Ru+3 = Ru+3_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.3	Unknown	lgK:1.2(2SF)	0	EES	
2	25	0.3	Unknown	lgK:2.17(3SF)	2	MSP	5398
3	25	1	HClO4	lgK:3.4(2SF)	2	MIS	852
4	25	0.1	HCl	lgR:1.4(2SF)	4	MSP	140
5	23	1	Ethanol:Water 40:60Vol HClO4	lgK:3.57(3SF)	5	MAG	5398

Reaction No. 5052							
Ru+3_Cl-1 + Cl-1 = Ru+3_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	6	0.21	HClO4	lgK:1.4(0.05SD)	5	MTP	5398
2	6	0.46	HClO4	lgK:1.3(0.05SD)	5	MTP	863
3	25	0.3	Unknown	lgK:0.4(1SF)	3	EES	5398
4	25	0.1	HCl	lgR:0.4(1SF)	4	MSP	140
5	23	1	Ethanol:Water 40:60Vol HClO4	lgK:3.14(3SF)	5	MAG	5398

Reaction No. 5053							
Ru+3_Cl-1(2) + Cl-1 = Ru+3_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	6	0.21	HClO4	lgK:0.5(0.05SD)	5	MTP	5398
2	6	0.46	HClO4	lgK:0.4(0.05SD)	5	MTP	863
3	25	0.1	HCl	lgR:0.4(0.05SD)	5	MSP	864

4	23	1	Ethanol:Water 40:60Vol HClO4	lgK:2.92 (3SF)	5	MAG	5398
---	----	---	------------------------------------	----------------	---	-----	------

Reaction No. 5054							
Ru+3_Cl-1(3) + Cl-1 = Ru+3_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	6	0.21	HClO4	lgK:-0.2 (0.05SD)	6	MTP	863
2	6	0.46	HClO4	lgK:-0.2 (0.05SD)	5	MTP	863
3	25	1	HClO4	lgK:2.4 (0.05SD)	4	MIS	852
4	23	1	Ethanol:Water 40:60Vol HClO4	lgK:2.64 (3SF)	5	MAG	5398

Reaction No. 5055							
Ru+4 + e-1 = Ru+3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.8 (3SF)	1	EDH	
2	25	0	Inf. Dilution	lgK:11.35 (4SF)	0	CRV	32224
3	25	0.5	HCl	lgK:15.4 (0.05SD)	5	MEF	866
4	25	2	HCl	lgK:14.5 (0.05SD)	5	MEF	866

Reaction No. 5056							
Cl-1 + Ru+4 = Ru+4_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:3.3 (0.05SD)	5	MIS	852

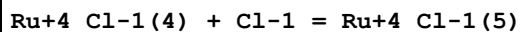
Reaction No. 5057							
Ru+4_Cl-1 + Cl-1 = Ru+4_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:2.7 (0.05SD)	5	MIS	852

Reaction No. 5058							
Ru+4_Cl-1(2) + Cl-1 = Ru+4_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:2.6 (0.05SD)	5	MIS	852

Reaction No. 5059							
Ru+4_Cl-1(3) + Cl-1 = Ru+4_Cl-1(4)							

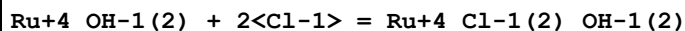
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:2.4 (0.05SD)	5	MIS	852

Reaction No. 5060



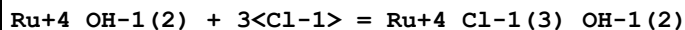
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:2.4 (0.05SD)	5	MIS	852

Reaction No. 5061



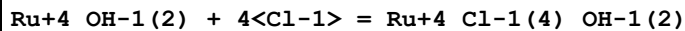
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	6	0.46	HClO4	lgK:1.1 (0.05SD)	5	MTP	863
2	6	0.92	HClO4	lgK:1.1 (0.05SD)	5	MTP	863
3	25	4	HClO4	lgK:1.4 (0.05SD)	5	MSP	851

Reaction No. 5062



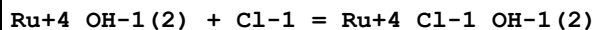
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	6	0.46	HClO4	lgK:0.8 (0.05SD)	5	MTP	863
2	6	0.92	HClO4	lgK:0.9 (0.05SD)	5	MTP	863
3	90	4	HClO4	lgK:6.0 (0.05SD)	0	MSP	5024

Reaction No. 5063



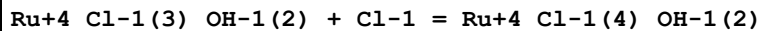
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	HClO4	lgK:2.2 (0.05SD)	5	MSP	851
2	90	4	HClO4	lgK:6.2 (0.05SD)	0	MSP	5024

Reaction No. 5064



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	6	0.46	HClO4	lgK:0.9 (0.05SD)	5	MTP	863
2	6	0.92	HClO4	lgK:0.9 (0.05SD)	5	MTP	863

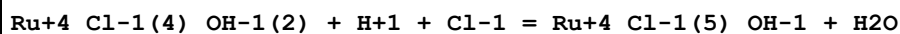
Reaction No. 5065



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
----	---	-------	--------	-------------	---	-------	--------

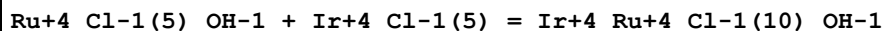
1	20	0.1:2.5	HCl	lgK:0.36(0.5SD)	3	MSP	848
2	25	4	HClO4	lgK:0.4(0.1SD)	0	MSP	848

Reaction No. 5066



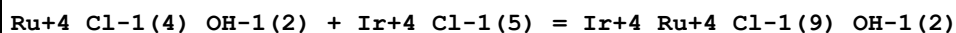
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	HClO4	lgR:0.2(0.05SD)	5	MSP	851

Reaction No. 5106



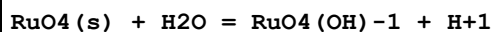
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.9(0.2SD)	3	MCO	850

Reaction No. 5107



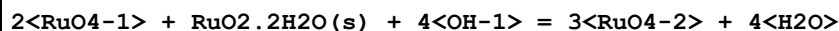
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.9(0.2SD)	3	MCO	850

Reaction No. 5109



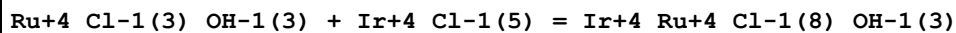
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.0(3SF)	1	ELC	

Reaction No. 5110



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.2(0.08SD)	5	MAN	858

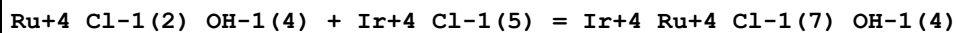
Reaction No. 5111



Note: Weighted out coz its thermodynamically isolated

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.9(0.2SD)	0	MCO	850

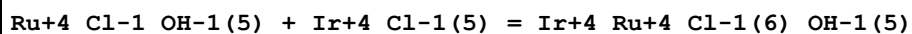
Reaction No. 5112



Note: Weighted out coz its thermodynamically isolated

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.9(0.2SD)	0	MCO	850

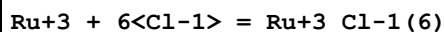
Reaction No. 5113



Note: Weighted out coz its thermodynamically isolated

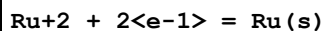
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.9(0.2SD)	0	MCO	850

Reaction No. 5117



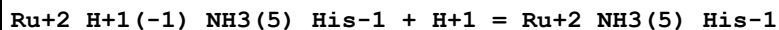
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:-4.(1SF)	0	MSP	931
2	25	10	Unknown	lgK:-4.(1.SD)	3	ENS	854

Reaction No. 5121



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.4(0.05SD)	4	ENS	802
2	25	0	Inf. Dilution	E:0.455(3SF)	5	CRV	6330

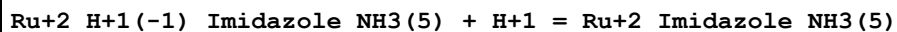
Reaction No. 29711



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.04:0.5	Unknown	lgK:8.8(2SF)	0	CMN	1388

Note (1): Thermodynamically isolated

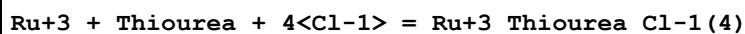
Reaction No. 29712



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.04:0.5	Unknown	lgK:9.0(2SF)	0	CMN	1388

Note (1): Thermodynamically isolated

Reaction No. 42821



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Wgt	lgK:18.92(4SF)	4	MSP	906

Reaction No. 42822							
Ru+3 + 2<Thiourea> + 3<Cl-1> = Ru+3_Thiourea(2)_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Wgt	lgK:22.72 (4SF)	4	MSP	906

Reaction No. 42823							
Ru+3 + 3<Thiourea> + 2<Cl-1> = Ru+3_Thiourea(3)_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Wgt	lgK:26.26 (4SF)	4	MSP	906

Reaction No. 44436							
Ru+3 + PhenThiourea + 4<Cl-1> = Ru+3_PhenThiourea_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Vol	lgK:18.23 (4SF)	4	MSP	906

Reaction No. 44437							
Ru+3 + 2<PhenThiourea> + 3<Cl-1> = Ru+3_PhenThiourea(2)_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Vol	lgK:21.62 (4SF)	4	MSP	906

Reaction No. 44438							
Ru+3 + 2<Cl-1> + 3<PhenThiourea> = Ru+3_PhenThiourea(3)_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Vol	lgK:24.74 (4SF)	4	MSP	906

Reaction No. 53004							
Cr+2 + Ru+3_Cl-1 = Ru+3_Cr+2_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:3.9 (2SF)	4	MKN	5398

Reaction No. 53036							
Ru+3_Cl-1(4) + Cl-1 = Ru+3_Cl-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:2. (0.5SD)	3	ESR	899

2	30	4	HClO ₄	lgK:0.9 (0.5SD)	3	ESR	899
3	23	1	Ethanol:Water 40:60Vol HClO ₄	lgK:2.31 (3SF)	5	MAG	5398

Reaction No. 54081							
SO ₄ -2 + Ru+2 = Ru+2_SO ₄ -2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	NaClO ₄	lgK:1.35 (3SF)	5	MCE	931
2	20:35	0	NaClO ₄	lgK:2.72 (3SF)	0	MCE	931
3	20:35	0.25	NaClO ₄	lgK:2.02 (3SF)	3	MCE	931
4	20:35	0.5	NaClO ₄	lgK:1.88 (3SF)	3	MCE	931
5	20:35	1	NaClO ₄	lgK:1.70 (3SF)	3	MCE	931
6	20:35	2	NaClO ₄	lgK:1.30 (3SF)	3	MCE	931
7	25	0	Inf. Dilution	lgK:3.219 (4SF)	1	EOR	
8	25	0	Inf. Dilution	lgK:2.36 (3SF)	3	EOD	22958
9	20:35	0:2	NaClO ₄	dH:0.0 (1SF)	4	MTD	931

Reaction No. 54082							
SO ₄ -2 + Ru+3 = Ru+3_SO ₄ -2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.95 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:1.95 (3SF)	5	EOD	22958
3	25	2	NaClO ₄	lgK:2.04 (3SF)	4	MPL	931

Reaction No. 54083							
2<SO ₄ -2> + Ru+3 = Ru+3_SO ₄ -2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.97 (4SF)	1	EOR	
2	25	2	NaClO ₄	lgK:3.57 (3SF)	4	MPL	931

Reaction No. 54202							
6<Cl-1> + Ru+2 = Ru+2_Cl-1 (6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:-13. (2SF)	0	MPL	931

Reaction No. 54203							
Ru+3_NH ₃ (5) + Cl-1 = Ru+3_NH ₃ (5)_Cl-1							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:1.63 (3SF)	3	MSP	140
2	25	0.11	NaClO ₄	lgK:1.85 (3SF)	4	MSP	931
3	35	0.1	Unknown	lgK:2.18 (3SF)	4	MAN	931
4	45	0.1	Unknown	lgK:2.21 (3SF)	4	MAN	931
5	54.7	0.25	NaTs	lgK:1.94 (3SF)	5	MKN	6780
6	64	0.1	Unknown	lgK:2.27 (3SF)	4	MAN	931
7	80	0.1	Unknown	lgK:2.32 (3SF)	4	MAN	931
8	90	0.1	Unknown	lgK:2.37 (3SF)	4	MAN	931

Reaction No. 54295							
Ru+3_NH3(5) + Br-1 = Ru+3_NH3(5)_Br-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:1.4 (2SF)	0	MSP	931
2	54.7	0.25	NaTs	lgK:1.63 (3SF)	5	MKN	6780

Reaction No. 54349							
Ru+3_NH3(5) + I-1 = Ru+3_NH3(5)_I-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:1.6 (2SF)	0	MSP	931
2	54.7	0.25	NaTs	lgK:1.80 (3SF)	5	MKN	6780

Reaction No. 54426							
RuO4-1 + MnO4-2 = RuO4-2 + MnO4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20			lgK:0.64 (2SF)	2	MSP	931
2	20			dH:-4.0 (2SF)	2	MCL	931

Reaction No. 54673							
Ru(s) + 2.5<O2(g)> + 0.5<H2(g)> = RuO4_OH-1 - e-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55.7 (3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-70. (5.SD)	0	ENS	869

Reaction No. 54674							
RuO4-2 + 2<H2O> + 2<e-1> = RuO2(s) + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:18.8 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:11.8 (0.1SD)	0	ENS	140

Reaction No. 54675							
RuO2(s) + 4<H+1> + 4<e-1> = Ru(s) + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.8 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:53.3 (0.1SD)	0	ENS	140
3	25	0	Inf. Dilution	E:0.68 (2SF)	0	CRV	22551
Note (3): Reaction in source is unbalanced; 2<H2O> assumed as product							

Reaction No. 54702							
RuO4 + e-1 = RuO4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.0 (0.05SD)	4	MIS	775
2	25	0	Inf. Dilution	E:0.99 (2SF)	4	CRV	22551

Reaction No. 54705							
Ru+4_C1-1(5) + Cl-1 = Ru+4_C1-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	H2SO4	lgK:3. (0.5SD)	3	ESR	899

Reaction No. 54706							
Ru+4 + 2<OH-1> = Ru+4_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.3 (3SF)	1	ELC	
2	25	1	Unknown	lgK:30. (3.SD)	0	ESR	899

Reaction No. 54738							
Ru(s) + 2<O2(g)> = RuO4(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-33. (1.SD)	3	ENS	869
2	25	0	Inf. Dilution	dH:-44. (1.SD)	4	ENS	869

Reaction No. 54739							
Ru+3_C1-1(5) + Cl-1 = Ru+3_C1-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0.1	HClO4	lgK:-0.2(0.3SD)	3	ESR	899
2	25	4	HClO4	lgK:-0.9(0.3SD)	3	ESR	899
3	35	4	NaClO4	lgK:-1.(0.3SD)	3	ESR	899

Reaction No. 54745							
Ru+3_NO + NO3-1 = Ru+3_NO_NO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	3	HNO3	lgR:-0.3(0.05SD)	5	MAN	5471

Reaction No. 54746							
Ru+3_NO + 2<NO3-1> = Ru+3_NO_NO3-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	3	HNO3	lgR:-1.0(0.05SD)	5	MAN	5471
2	20	4.5	HNO3	lgR:-0.6(0.05SD)	5	MIX	5041

Reaction No. 54747							
Ru+3_NO + 3<NO3-1> = Ru+3_NO_NO3-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	3	HNO3	lgR:-0.7(0.05SD)	5	MAN	5471
2	20	4.5	HNO3	lgR:-0.7(0.05SD)	5	MIX	5041

Reaction No. 54748							
Ru+3_NO_Cl-1(4) + H2O = H+1 + Ru+3_NO_Cl-1(4)_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:-6.0(0.2SD)	3	MGL	849

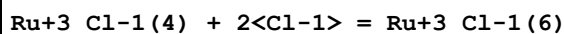
Reaction No. 54749							
Ru+3_NO_Cl-1(3) + H2O = H+1 + Ru+3_NO_Cl-1(3)_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:-5.0(0.2SD)	3	MGL	849

Reaction No. 54750							
Ru+3_NO_Cl-1(3)_OH-1 + H2O = H+1 + Ru+3_NO_Cl-1(3)_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:-7.5(0.2SD)	3	MGL	849

Reaction No. 54752							
Ru+3 + NO = Ru+3_NO							

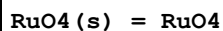
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0:100	0:10	Unknown	lgK:100.(30.SD)	3	EES	

Reaction No. 54780



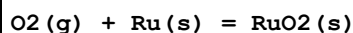
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4:6	HCl	lgK:-1.3(0.1SD)	4	MSP	5473

Reaction No. 57058



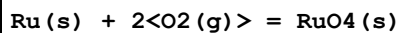
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.9(0.05SD)	5	ENS	4998

Reaction No. 57755



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:44.27(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:43.94(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:30.68(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:22.76(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:17.52(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-61.(1.SD)	0	ENS	869
7	25	0	Inf. Dilution	dG:-60.39(4SF)	3	EFE	6422
8	25	0	Inf. Dilution	dG:-60.394(5SF)	3	CRV	31291
9	25	0	Inf. Dilution	dH:-73.(2.SD)	2	ENS	869
10	25	0	Inf. Dilution	dH:-72.9(3SF)	2	MTD	6422
11	127	0	Inf. Dilution	dH:-72.7(3SF)	1	MTD	6422
12	227	0	Inf. Dilution	dH:-72.2(3SF)	0	MTD	6422
13	327	0	Inf. Dilution	dH:-71.7(3SF)	0	MTD	6422

Reaction No. 57756



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-36.(1.SD)	0	ENS	869
3	25	0	Inf. Dilution	dH:-57.(2.SD)	2	ENS	869

Reaction No. 57757							
$\text{Ru(s)} + 2\text{O}_2(\text{g}) = \text{RuO}_4$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-35.(1.SD)	0	ENS	869
3	25	0	Inf. Dilution	dH:-57.(2.SD)	2	ENS	869

Reaction No. 57758							
$\text{Ru(s)} + 2\text{O}_2(\text{g}) = \text{RuO}_4\text{-1} - \text{e-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-58.(1.SD)	0	ENS	869

Reaction No. 57759							
$2\text{O}_2(\text{g}) + \text{Ru(s)} + 2\text{e-1} = \text{RuO}_4\text{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-72.(1.SD)	3	ENS	869
2	25	0	Inf. Dilution	dG:-71.600(5SF)	4	CRV	9029
3	25	0	Inf. Dilution	dG:-72.(2SF)	3	EFE	30708
4	25	0	Inf. Dilution	dH:-110.200(6SF)	4	CRV	9029
5	25	0	Inf. Dilution	dH:-109.4(4SF)	3	EOD	30708

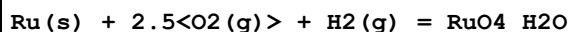
Reaction No. 57798							
$\text{Ru(s)} = \text{Ru(g)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:144.(1.SD)	3	ENS	869
2	25	0	Inf. Dilution	dH:155.(1.SD)	3	ENS	869

Reaction No. 57799							
$\text{Ru(s)} + 2\text{H}_2(\text{g}) + 2\text{O}_2(\text{g}) = \text{RuO}_2.2\text{H}_2\text{O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:120.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-51.(1.SD)	0	ENS	869

Reaction No. 57800							
$\text{Ru(s)} + 2\text{O}_2(\text{g}) + 0.5\text{H}_2(\text{g}) = \text{H+1_RuO}_4\text{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

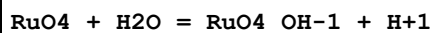
1	25	0	Inf. Dilution	dG:-35.(1.SD)	3	ENS	869
---	----	---	---------------	---------------	---	-----	-----

Reaction No. 57801



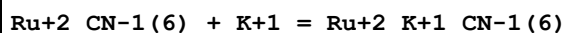
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-80.(5.SD)	3	ENS	869

Reaction No. 57802



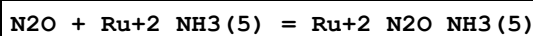
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-11.4(0.1SD)	5	ENS	4998
2	25	0	Inf. Dilution	lgK:-11.9(0.1SD)	5	ENS	4998

Reaction No. 57807



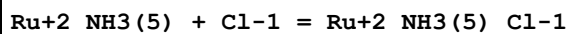
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.5(0.05SD)	5	MCN	5398

Reaction No. 57812



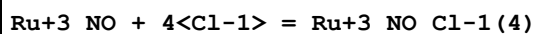
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	6.8	0.023	HCl	lgK:0.9(0.05SD)	5	MKN	5398
2	20.1	0.023	HCl	lgK:0.7(0.05SD)	5	MKN	5398

Reaction No. 57815



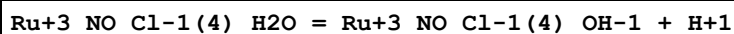
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	HClO4	lgK:0.2(0.05SD)	5	MSP	5398

Reaction No. 57816



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	55	0.5	NaClO4	lgK:0.6(0.05SD)	5	MAX	5398
2	55	0.5	NaClO4	lgK:0.6(0.05SD)	5	MSP	5398

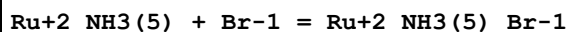
Reaction No. 57817



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
----	---	-------	--------	-------------	---	-------	--------

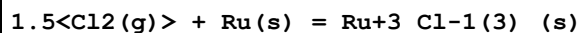
1	50.5	0.5	NaClO4	lgK:-5.3(0.05SD)	4	MKN	5398
---	------	-----	--------	------------------	---	-----	------

Reaction No. 57827



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	HClO4	lgK:0.04(0.01SD)	5	MSP	5398

Reaction No. 67224



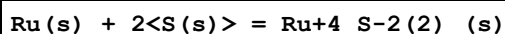
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.98(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:27.73(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:17.76(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:11.85(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:7.960(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-38.17(4SF)	3	EFE	6422
7	25	0	Inf. Dilution	dH:-205.0(4SF)kJ	3	CRV	6330
8	25	0	Inf. Dilution	dH:-55.0(3SF)	2	MTD	6422
9	127	0	Inf. Dilution	dH:-54.44(4SF)	2	MTD	6422
10	227	0	Inf. Dilution	dH:-53.72(4SF)	2	MTD	6422
11	327	0	Inf. Dilution	dH:-52.90(4SF)	0	MTD	6422

Reaction No. 67225



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:136.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:135.9(4SF)	4	ENS	6422
3	85	0	Inf. Dilution	lgK:110.8(4SF)	4	ENS	6422
4	25	0	Inf. Dilution	dG:-186.7(4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dH:-213.4(4SF)	4	MTD	6422
6	85	0	Inf. Dilution	dH:-212.6(4SF)	4	MTD	6422

Reaction No. 67226



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.07(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:33.84(4SF)	0	ENS	6422

3	127	0	Inf. Dilution	lgK:24.85 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:19.30 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:15.55 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-46.47 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-49.20 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-50.41 (4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-51.20 (4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-51.79 (4SF)	0	MTD	6422

Reaction No. 67227							
Ru(s) + 2<Se(s)> = Ru+4_Se-2(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.65 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:26.48 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:19.44 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:15.19 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:12.14 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-36.36 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-38.60 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-38.69 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-41.80 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-42.07 (4SF)	4	MTD	6422

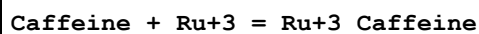
Reaction No. 67524							
Ru+2_Pyrazine_CN-1(5) + H+1 = Ru+2_H+1_Pyrazine_CN-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:0.4 (0.1SD)	0	MPS	6530
2	25	0	Inf. Dilution	lgK:0.4 (1SF)	1	ESC	

Reaction No. 69745							
Theophylline-1 + Ru+3 = Ru+3_Theophylline-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	LiClO4	lgK:3.73 (3SF)	4	MGL	4554
2	25	1	NaClO4	lgK:1.16 (3SF)	0	MGL	2261

Reaction No. 69746							
Theophylline-1 + Ru+2 = Ru+2_Theophylline-1							

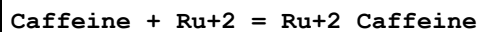
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	LiClO4	lgK:9.4 (2SF)	3	MGL	4554

Reaction No. 69747



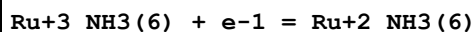
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	LiClO4	lgK:3.20 (3SF)	4	MGL	4554
2	25	1	NaClO4	lgK:1.19 (3SF)	0	MGL	2261

Reaction No. 69748



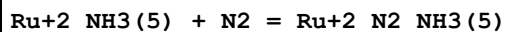
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	LiClO4	lgK:9.4 (2SF)	3	MGL	4554

Reaction No. 70256



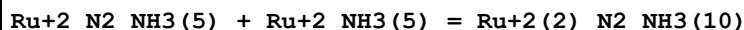
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.10 (2SF)	5	CRV	6330
2	25	0	Inf. Dilution	E:0.24 (2SF)	0	ENS	7276

Reaction No. 70257



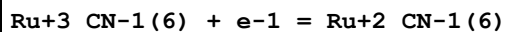
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.48 (3SF)	0	ENS	7276

Reaction No. 70258



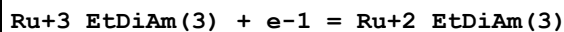
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.85 (3SF)	0	ENS	7276

Reaction No. 71020



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.86 (2SF)	5	CRV	6330

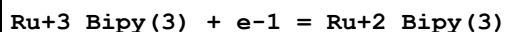
Reaction No. 71021



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
----	---	-------	--------	-------------	---	-------	--------

1	25	0	Inf. Dilution	E:0.210 (3SF)	5	CRV	6330
---	----	---	---------------	---------------	---	-----	------

Reaction No. 71022



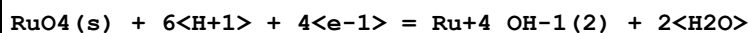
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.24 (3SF)	5	CRV	6330

Reaction No. 71023



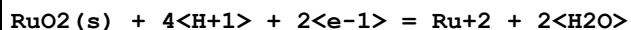
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:139.5 (4SF)	1	ELC	
2	25	0	Inf. Dilution	E:1.038 (4SF)	0	CRV	6330

Reaction No. 71024



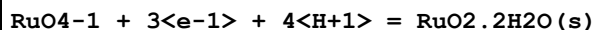
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:98.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:1.40 (3SF)	0	CRV	6330

Reaction No. 71025



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:1.120 (4SF)	0	CRV	6330

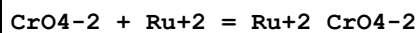
Reaction No. 74285



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.533 (4SF)	1	CRV	22551

Note (1): Phase unclear; given as RuO2(s,hydr)

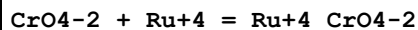
Reaction No. 75011



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0.01	0	Inf. Dilution	lgK:2.8048 (5SF)	3	EOD	22958
2	25	0	Inf. Dilution	lgK:2.6212 (5SF)	3	EOD	22958
3	60	0	Inf. Dilution	lgK:2.5048 (5SF)	3	EOD	22958

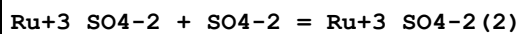
4	100	0	Inf. Dilution	lgK:2.4927 (5SF)	3	EOD	22958
5	150	0	Inf. Dilution	lgK:2.6259 (5SF)	3	EOD	22958
6	200	0	Inf. Dilution	lgK:2.9312 (5SF)	3	EOD	22958

Reaction No. 75035



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0.01	0	Inf. Dilution	lgK:5.0069 (5SF)	3	EOD	22958
2	25	0	Inf. Dilution	lgK:5.6566 (5SF)	3	EOD	22958
3	60	0	Inf. Dilution	lgK:6.6159 (5SF)	3	EOD	22958
4	100	0	Inf. Dilution	lgK:7.6950 (5SF)	3	EOD	22958
5	150	0	Inf. Dilution	lgK:9.0221 (5SF)	3	EOD	22958
6	200	0	Inf. Dilution	lgK:10.3770 (6SF)	3	EOD	22958

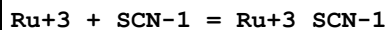
Reaction No. 76124



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:1.53 (3SF)	5	MVM	11516

Note (1): Lazarev L et al., Zhur. Neorg. Khim., 1968, 13, 598

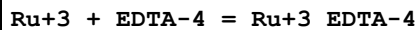
Reaction No. 76149



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	70	1	NaClO4	lgK:1.78 (3SF)	5	MSP	11516

Note (1): Yaffee R et al., J. Am. Chem. Soc., 1952, 72, 2500

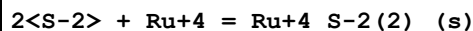
Reaction No. 76223



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.1 (3SF)	1	ESC	
2	30	0.1	KCl	lgK:13.8 (3SF)	5	MGL	11516

Note (2): Taqui-Khan M et al., Indian J. Chem., 1982, 21A, 441

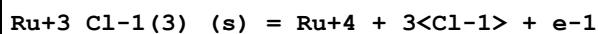
Reaction No. 78992



Note: The species S-2 does not exist in aqueous solution See reaction: $\text{H+1} + \text{S-2} = \text{H+1_S-2}$

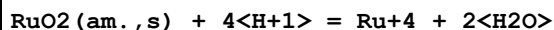
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:100.8 (4SF)	0	ELC	

Reaction No. 81565



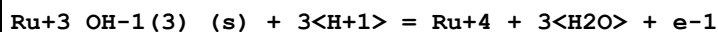
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.7 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-6.39 (3SF)	0	CRV	32224

Reaction No. 81566



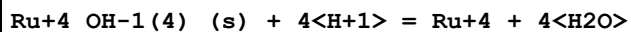
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.51 (3SF)	4	CRV	32224

Reaction No. 81567



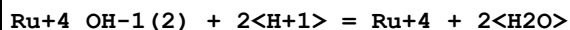
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.95 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-3.55 (3SF)	0	CRV	32224

Reaction No. 81568



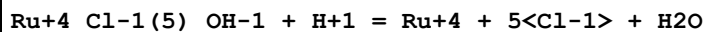
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.30 (4SF)	4	CRV	32224

Reaction No. 81569



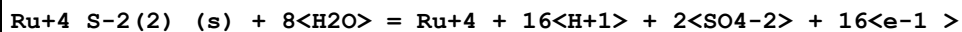
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.73 (3SF)	4	CRV	32224

Reaction No. 81570



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.13 (3SF)	4	CRV	32224

Reaction No. 81571



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
----	---	-------	--------	-------------	---	-------	--------

1	25	0	Inf. Dilution	lgK:-141.8 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-144.86 (5SF)	0	CRV	32224

Reaction No. 81572							
RuO4(s) + 8<H+1> + 4<e-1> = Ru+4 + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:103.1 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:100.55 (5SF)	0	CRV	32224

Reaction No. 81573							
RuO4-1 + 8<H+1> + 3<e-1> = Ru+4 + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:87.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:82.20 (4SF)	0	CRV	32224

Reaction No. 81574							
RuO4 + 8<H+1> + 4<e-1> = Ru+4 + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:104.0 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:99.43 (4SF)	0	CRV	32224

Reaction No. 81575							
RuO4-2 + 8<H+1> + 2<e-1> = Ru+4 + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:77.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:72.01 (4SF)	0	CRV	32224

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt

3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CMN	Several experimental values (crit.)
CRV	Critical review
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
ESC	Standard conditions anchor
ESR	Similar reaction(s)
MAG	Silver electrode e.m.f.
MAN	Chemical analysis
MAX	Anion exchange
MCE	Cation exchange
MCL	Calorimetry
MCN	Conductivity
MCO	Colorimetry (often for measuring pH)
MEF	Electromotoric force, not specified
MGL	Glass electrode
MIS	Ion selective electrode
MIX	Ion exchange
MKN	Rate of reaction
MPL	Polarography
MPS	Potentiometry and Spectrophotometry
MSL	Solubility

MSP	Spectroscopy
MTD	Temperature difference
MTP	Migration or transference number
MVM	Voltametric Methods

References

Reference No.: 140(64SiM)

Stability Constants of Metal-ion Complexes SP17, 1964 Sillen LG; Martell AE; The Chemical Soc., London

Reference No.: 775(71CCM)

Selected Constants, 1971 Charlot G; Collumeau A; Marchon MJC; Oxidation-Reduction Potentials of Inorganic Substances in Aqueous Solution; Butterworths, London

Reference No.: 802(90Wea)

CRC Handbook of Chemistry and Physics, 70th Edn., 1990 Weast RC; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio

Reference No.: 848(74SPK)

Izv. Akad. Nauk SSSR, Ser. Khim., 1974, 23, 268 - 271; Shlenskaya VI; Pichugina GV; Khvostova VP; Alimarin IP; Spectrophotometric Study of Equilibriums in Ruthenium(IV) Chloride Solutions

Reference No.: 849(64MCW)

Inorg. Chem., 1964, 3, 1018 - 1024; Mercer EE; Campbell WM Jr; Wallace RM; Chloro Complexes of Nitrosylruthenium

Reference No.: 850(71KaY)

Russ. J. Inorg. Chem., 1971, 16, 997 Kalinina VE; Yatsimirskii KB; Joint Catalytic Effect of Ruthenium(IV) and Iridium(IV) Chloro Compounds in the Oxidation of Iodide by Periodate

Reference No.: 851(71NeV)

Dokl. Bolg. Akad. Nauk, 1971, 24, 1489 Neikova EA; Vasilev Kh; Preparation of Solutions of Ruthenium(IV) Complexes containing Chlorine by reducing Ruthenium(VI) of Ruthenates

Reference No.: 852(71PSS)

Russ. J. Inorg. Chem., 1971, 16, 1349 Pilipenko AT; Sereda IP; Semchinskaya ZA; Stability of Chloro-Complexes of Ruthenium

Reference No.: 853(49DwB)

J. Proc. R. Soc. N.S.W., 1949, 83, 146 - 149; Dwyer FP; Backhouse JR; Chemistry of Ruthenium. V. Potential of the Bivalent/Trivalent Ruthenium couple in Hydrochloric Acid

Reference No.: 854(66BuM)

J. Phys. Chem., 1966, 70, 3103 Buckley RR; Mercer EE; The Potential of the Ruthenium(II) - Ruthenium(III) Couple

Reference No.: 858(52CoH)

J. Am. Chem. Soc., 1952, 74, 5012 - 5015; Connick RE; Hurley CR; Chemistry of Ru(VI), - (VII), and -(VIII). Reactions, Oxidation Potentials and Spectra

Reference No.: 859(67BrL)

Arkiv Kemi, 1967, 26, 401 Brito F; Lewis D; Studies on the Chemistry of Ruthenium. 3. Reduction in 1M Sulfate Media

Reference No.: 860(57StK)

Zh. Neorg. Khim., 1957, 2, 444 - 451; Starik IE; Kositsyn AV; The State of Small Quantities of the Radioactive Elements in Solutions. I. The State of Tri- and Quadrivalent Ruthenium in Hydrochloric Acid Solutions

Reference No.: 861(57StK)

Zh. Neorg. Khim., 1957, 2, 1171 - 290; Starik IE; Kositzyn AV; The State of Small Quantities of Radioelements in Solutions. II. The State of Thallium in Sulfuric Acid Solutions

Reference No.: 863(700hO)

Radiochim. Acta, 1970, 13, 10 - 15; Ohyoshi E; Ohyoshi A; Studies on Ruthenium Chlorocomplexes by Paper Electrophoresis

Reference No.: 864(61CoF)

J. Am. Chem. Soc., 1961, 83, 3414 - 3416; Connick RE; Fine DA; The Identification of Neutral Ruthenium(III) Chloride Complexes: Equilibria Involving Neutral and Cationic Species

Reference No.: 866(41GrF)

Z. Elektrochem., 1941, 47, 208 - 211; Grube G; Fromm G; Oxidation Stages of Ruthenium. II. The Oxidation Potential of Ru(III) and Ru(IV) in Hydrochloric Acid Solution

Reference No.: 867(62AtD)

J. Am. Chem. Soc., 1962, 84, 2659 - 2661; Atwood DK; DeVries T; The Electrode Potential of Ruthenium(IV) and Its Lower Oxidation States

Reference No.: 869(68GoH)

Chem. Rev., 1968, 68, 229 - 252; Goldberg RN; Hepler LG; Thermochemistry and Oxidation Potentials of the Platinum Group Metals and their Compounds

Reference No.: 899(90Mur)

Estimation of Constants for JESS Database, 1990 Murray K; Record of ad hoc estimations

Reference No.: 906(79Per)

IUPAC Chemical Data Series No. 22, 1979 Perrin DD; Stability Constants of Metal-Ion Complexes, First Edition Part B, Organic Ligands; Pergamon, Oxford

Reference No.: 931(71SiM)

Stability Constants of Metal-ion Complexes SP25, 1971 Sillen LG; Martell AE; Supplement No 1 to Special Publication No 25; Chemical Society, London

Reference No.: 1388(74SuM)

Chem. Rev., 1974, 74, 471 - 517; Sundberg RJ; Martin B; Interactions of Histidine and Other Imidazole Derivatives with Transition Metal Ions in Chemical and Biological Systems

Reference No.: 2261(93PeP)

IUPAC Stability Constants Database, 1993 Pettit LD; Powell HKJ; IUPAC (in co-operation with Academic Software), Otley, UK

Reference No.: 4554(75ClT)

J. Am. Chem. Soc., 1975, 97, 1397 - 1403; Clarke MJ; Taube H; Nitrogen-Bound and Carbon-Bound Xanthine Complexes of Ruthenium Ammines

Reference No.: 4998(86BaM)

The Hydrolysis of Cations, 1986 Baes CF Jr; Mesmer RE; Robert E. Krieger Publ. Co. Malabar, Florida

Reference No.: 5024(72SBK)

Russ. J. Inorg. Chem., 1972, 17, 572 Shlenskaya VI; Biryukov AA; Kadomtseva VM; Ruthenium(IV) Chloride Complexes in Solutions

Reference No.: 5041(59FBG)

J. Inorg. Nucl. Chem., 1959, 12, 154 Fletcher JM; Brown PGM; Gardner ER; Hardy CJ; Wain AG; Woodhead JL; Nitrosylruthenium Nitrate Complexes in Aqueous Nitric Acid

Reference No.: 5398(82Hog)

IUPAC Chemical Data Series No. 21, 1982 Hogfeldt E; Stability Constants of Metal-ion Complexes, 2nd Suppl., Part A, Inorganic Ligands; Pergamon, Oxford

Reference No.: 5471(57BFW)

AERE C/R2260, 1957 Brown PGM; Fletcher JM; Wain AG

Reference No.: 5473(73SKT)

Vestn. Mosk. Univ., Khim., 1973, 28, 122 - 124; Shlenskaya VI; Kuratashvili ZA; Tikhonov IG; Spectrophotometric investigation of the Aquation and Anation of $K_2[RuCl_5H_2O]$

Reference No.: 6330(95LiF)

CRC Handbook of Chemistry and Physics, 76th Edn., 1995 Lide DR; Frederikse HPR; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio, U.S.A.

Reference No.: 6422(95BaP)

Thermochemical Data of Pure Substances, 3rd Edn., 1995 Barin I; Platzki G; Weinheim, Germany

Reference No.: 6530(83JoS)

Inorg. Chem., 1983, 22, 1117 - 1123; Johnson CR; Shepherd RE; The pKa of Pyraziniumpentacyanoruthenate(II), $(CN)_5Ru(pzH)^{-2}$

Reference No.: 6780(71BrK)

Inorg. Chem., 1971, 10, 85 - 87; Broomhead JA; Kane-Maguire L; Substitution Reactions of Ruthenium(III)-Ethylenediamine and Related Complexes. III. Halide Anation of cis-Diaquobis(ethylene-diamine)-, cis-Halogenoaquobis(ethylenediamine)-. and Aquopentaammineruthenium(III) Complexes

Reference No.: 7276(88CoW)

Advanced Inorganic Chemistry, 5th Edn., 1988 Cotton FA; Wilkinson G; John Wiley & Sons Inc., N.Y., U.S.A

Reference No.: 9029(97SSW)

Geochim. Cosmochim. Acta, 1997, 61, 907 - 950; Shock EL; Sassani DC; Willis M; Sverjensky DA; Inorganic Species in Geologic Fluids: Correlations among Standard Molal Thermodynamic Properties of Aqueous Ions and Hydroxide Complexes

Reference No.: 11516(00PeP)

IUPAC Stability Constants Database, 2000 Pettit LD; Powell HKJ; Release 4; Academic Software, U.K.

Reference No.: 22551(06Inz)

Encyclopedia of Electrochemistry, Vol. 7a, 2006, 17 - 75; Inzelt G; Standard, Formal and Other Characteristic Potentials of Selected Electrode Reactions

Reference No.: 22958(10AML)

Appl. Geochem., 2010, 25, 242 - 260; Accornero M; Marini L; Lelli M; Prediction of the Thermodynamic Properties of Metal-chromate Aqueous Complexes to High Temperatures and Pressures and Implications for the Speciation of Hexavalent Chromium in Some Natural Waters

Reference No.: 30708(69Hep)

Can. J. Chem., 1969, 47, 3469 - 3470; Hepler LG; Thermodynamic Properties of Ruthenate and Molybdate ions

Reference No.: 31291(84PSG)

U.S. Bur. Mines Bull. 677, 1984 Pankratz LB; Stuve JM; Gokcen NA; Thermodynamic Data for Mineral Technology; US Det. Interior, Bur. Mines, Washington, DC, USA

Reference No.: 32224(80BeT)

Lawrence Berkeley Laboratory Report LBL-11448, 1980 Benson LV; Teague LS; A Tabulation of Thermodynamic Data for Chemical Reactions Involving 8 Elements Common to Radioactive Waste Package Systems